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1) Define cloud computing and explain any four characteristics

Definition: Cloud Computing is the delivery of computing services such as servers, storage, databases, networking, and software over the internet.

Four characteristics:

1) On-demand self-service: users can access computing resources whenever needed without contacting the service provider.

2) Broad network access: Cloud services can be accessed through the internet using computers, laptops, and mobile phones.

3) Resource pooling: Computing resources are shared among multiple users.

4) Rapid elasticity: Resource can be increased or decreased quickly according to demand.

Difference: Traditional on-premise computing requires organizations to purchase and maintain their own hardware, whereas cloud computing provides resources over the internet with flexible usage.

2) Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

1) mainframe computers: Large computers served multiple users through terminals.

2) personal computers: computing becomes aviable to individuals and organizations.

3) client-server computing: clients requested services from centralized servers.

4) Distributed computing: multiple computers worked together to perform tasks.

5) virtualization: one physical server could run multiple virtual machines.

6) Internet and web service: Applications and resources became accessible through the internet.

7) modern cloud computing: providers began offering computing resources, platforms, and software on demand.

Conclusion: Advances in hardware, networking, virtualization, and internet software enabled scalable, flexible, and cost-effective cloud platforms.

Q.5 Explain the role of server virtualization in cloud computing. Differentiable between virtualization as an enabling technology and cloud computing as a service model.

Server virtulization: Server virtualization is a technology that divides one physical server into multiple virtual machines (VMs). Each VM can run its own operating system and applications.

Role in cloud computing:

→ improves hardware utilization.
→ Reduces hardware costs.
→ Allowed multiple operating system on one server
→ Flexibility and easy
→ Support quick creation.

Difference:

Virtualization	cloud computing.
A technology that creats virtual resources	A service model that delivers computing resources over the internet.
Runs multiple VMs on physical hardware	provides servers, storage, Platforms, and software.
Eg: VM ware workstation.	Eg: Amazon Web (AWS).

Con: Virtualization is an enabling technology, while cloud computing delivers computing resurces as services.

4) Discuss the role of data centers in delivering cloud service.

Def: A data center is a facility containing servers, storage devices, networking equipment, and supporting infrastructure used to store, process, and manage data.

Roles of data centers:

1) Data storage: Stores large amounts of user and business data.

2) Data processing: Runs application and processes user requests.

3) Networking: Connects servers to the internet and users.

4) Security: Uses access controls, monitoring, and other security measures to protect data.

5) Availability: Uses backup systems and redundant equipment to reduce service interruptions.

6) Scalability: Provides additional computing resources when demand increases.

Eg: Google Drive, Gmail, and Google cloud.

Con: Data centers are the physical foundation of cloud computing because they provide the infrastructure required to deliver reliable cloud services.

Differentiate Iaas, Paas, Saas, and Xaas with one suitable real-word example for each model.

Service Model	Meaning	What it provides	Example.
Iaas	Infrastructure as a service	Virtual machines, storage, and networking	Amazon EC2
Paas	Platform as a service	A platform to develop, test, and deploy application	Google App Engine.
Saas	Software as a service	Ready-to-use software over the internet.	Google Docs.
Xaas	Anything as a service	A broad range of IT services delivered as services	Microsoft Azure services.

Explanation:

→ Iaas: Users manage their operating system and applications, while the provider manges the underlying infrastructure.

→ Paas: Developers build applications with managing.

→ Saas: Users access complete software through a browser or application.

→ Xaas: many types of resources and capabilities are provided servies over a network.